BUSINESS INTELLIGENCE LAB

UNIT 3 | SESSION 9

EXPERIMENT 9

BI Security: Authentication, Authorization and Access Control using Microsoft Power BI

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1. AIM

To understand and implement Business Intelligence security concepts including authentication, authorization, role-based access control, and data protection mechanisms using Microsoft Power BI.

2. OBJECTIVES

- To identify organizational users of the Global Superstore BI system and their information requirements.
- To prepare a Role Matrix mapping each user to the reports and dashboards they need.
- To study and compare authentication methods used in Power BI.
- To design an Access Control Matrix using View, Edit, Share and Admin permissions.
- To create a secure Power BI workspace and assign workspace roles.
- To design and implement Row-Level Security (RLS) so that each Regional Manager sees only their own regions.
- To test RLS in Power BI Desktop and in the Power BI Service.
- To assess security risks and prepare security recommendations.

3. TOOLS / SOFTWARE REQUIRED

1. Microsoft Power BI Desktop (Manage roles, View as, Publish).
2. Microsoft Power BI Service (workspaces, Manage access, Row-Level Security, Test as role).
3. Global Superstore Dataset.xlsx.
4. Microsoft Excel for the Role, Permission and Recommendation matrices.
5. DAX (Data Analysis Expressions) for RLS filter rules.

4. DATASET USED

Dataset: Global Superstore Dataset (Kaggle), using the same Power BI data model as Experiments 5 to 8, saved and published as **BI Expe-9**.

The data model contains the following tables:

Table	Role in the model	Security relevance
Fact_Sales	Sales transactions: Order ID, Region, Sales, Profit, Quantity, Discount	RLS filters are applied here on the Region column

Table	Role in the model	Security relevance
Dim_Customer	Customer details	Contains customer names (personal data)
Dim_Product	Product, Category, Sub-Category	Low sensitivity
Dim_Region	Region and geography	Basis for regional access
Dim_Date	Calendar for time analysis	Low sensitivity
Orders, People, Returns	Source order data, regional managers list, returned orders	People table maps regions to managers

The Region field contains 23 regions (for example Western Europe, Central US, Eastern Asia, North Africa). These are grouped into five management areas for regional access control.

5. THEORY

5.1 Security in Business Intelligence

Security is a critical component of Business Intelligence systems. BI platforms often contain sensitive organizational information, such as revenue, profit and customer data, that must be protected from unauthorized access. A secure BI system answers three questions: who the user is (authentication), what the user is allowed to do (authorization), and which data the user is allowed to see (access control).

5.2 Authentication

Authentication verifies the identity of a user. Power BI uses Microsoft Entra ID (formerly Azure Active Directory) organizational accounts. Users sign in with a work or school account, and personal email accounts cannot be used to sign up for the Power BI Service. Organizations can strengthen sign-in with Multi-Factor Authentication (MFA) and Conditional Access policies.

5.3 Authorization

Authorization determines the resources and actions a user is permitted to access after signing in. In Power BI this is controlled through workspace roles, item sharing permissions, and app audiences.

5.4 Workspace Roles in Power BI

Role	Permissions
Viewer	View and interact with reports and dashboards; cannot edit or share content.
Contributor	Viewer rights plus create, edit, publish and delete content in the workspace.
Member	Contributor rights plus share content, publish apps and add users with lower roles.
Admin	Full control: all Member rights plus manage all access, update and delete the workspace.

5.5 Row-Level Security (RLS)

Row-Level Security restricts data at row level for a given user. Roles are defined in Power BI Desktop with DAX filter expressions that return TRUE or FALSE for each row, for example [Region] = "Western Europe". After publishing, users or security groups are assigned to roles in the Power BI Service. Two approaches are used:

- **Static RLS:** each role has a fixed filter, such as a list of regions.
- **Dynamic RLS:** one role uses USERPRINCIPALNAME() to look up the signed-in user's regions in a security mapping table.

Important: RLS applies only to users with the **Viewer** role or users given read access through sharing or apps. Workspace Admins, Members and Contributors can see all data regardless of RLS.

5.6 Business Scenario

Global Superstore has multiple departments including Sales, Finance, and Regional Management. Each department requires access only to information relevant to its responsibilities. Acting as BI Security Administrators, a secure access strategy is designed and implemented for the organizational reports and dashboards.

6. PROCEDURE

Part A: User and Role Identification

Step 1: Identified organizational users of the Global Superstore BI system.

Step 2: Defined information requirements for each user category.

User	Department	Information requirement	Data scope
CEO	Executive	Overall KPIs, trends, scorecard and insights for strategic decisions	All regions
Finance Manager	Finance	Sales, Profit, Profit Margin, targets and achievement	All regions
Sales Manager	Sales	Sales, Orders, customers, products and sales trends	All regions
Regional Manager (5)	Regional Management	Sales and profit performance of their own regions only	Own regions only
BI Developer / Analyst	IT / BI	Builds and maintains reports and the data model	All (development)
BI Administrator	IT / BI	Manages workspaces, access, roles and security	All (administration)

Step 3: Created the Role Matrix showing which report pages of the BI Expe-9 report each user should access.

Report page	CEO	Finance Mgr	Sales Mgr	Regional Mgr	BI Dev / Admin
Geographical Sales Analysis	✓	✓	✓	✓ (own region)	✓
Relational Report (customer and order detail)	✓	✗	✓	✗	✓
Matrix Report	✓	✓	✓	✓ (own region)	✓
Comparative Analysis	✓	✓	✓	✗	✓
Dashboard (Interactive)	✓	✓	✓	✓ (own region)	✓
Executive Dashboard	✓	✓	✓	✓ (own region)	✓
Business Scorecard	✓	✓	✗	✗	✓
Business Insights	✓	✓	✓	✗	✓

The Relational Report shows customer names and order IDs, so it is limited to roles that need customer-level detail. Regional Managers receive only regional pages, and every page they open is filtered by RLS.

Part B: Authentication Analysis

Step 4: Studied authentication in Power BI. Power BI Desktop and the Power BI Service both require sign-in with a Microsoft Entra ID organizational account. In this experiment, sign-in used the Chandigarh University organizational account (24BAI70236), which is managed centrally by the university tenant.

Step 5: Compared authentication methods.

Criteria	Username-Password	Organizational Account (Entra ID)	Multi-Factor Authentication
How it works	User enters a password only	Single identity managed centrally by the organization	Password plus a second factor (authenticator app, SMS, call)
Security level	Low	Medium to high	High
Main weakness	Passwords can be guessed, reused or phished	Still vulnerable if the password is stolen and no second factor is used	Slightly more effort for users at sign-in
Central control	No	Yes: accounts can be disabled instantly when staff leave	Yes, enforced through Conditional Access
Use in Power BI	Not supported for personal accounts	Required to use Power BI Service	Recommended for all users, mandatory for admins

Step 6: Documented the advantages of secure authentication.

- **Verified identity:** only known organizational users can reach BI content.
- **Single sign-on:** one account for Power BI, Excel, Teams and Outlook reduces password reuse.
- **Central control:** IT can disable a leaving employee's account and all access ends at once.
- **MFA protection:** a stolen password alone is not enough to access sensitive financial data.
- **Accountability:** every action is linked to a named user, which supports audit logs.

Part C: Authorization and Access Control Design

Step 7: Created a dedicated workspace for secure sharing. My workspace cannot be shared with others, so a new workspace named **Global Superstore BI** was created in the Power BI Service (Workspaces, then New workspace), with the student account as Owner.

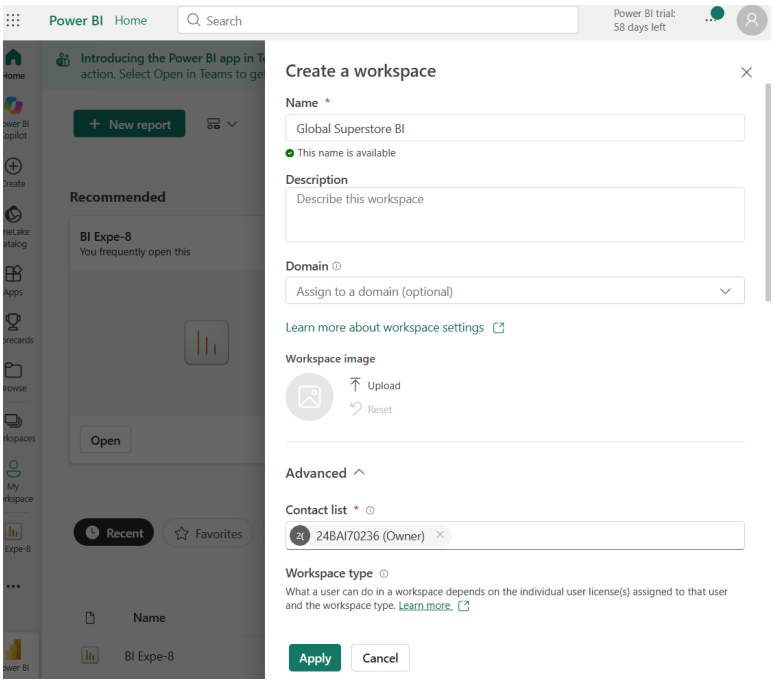


Figure 1: Creating the Global Superstore BI workspace in the Power BI Service.

The report was then published from Power BI Desktop to this workspace (Home, Publish, Global Superstore BI), which created the BI Expe-9 report and semantic model.

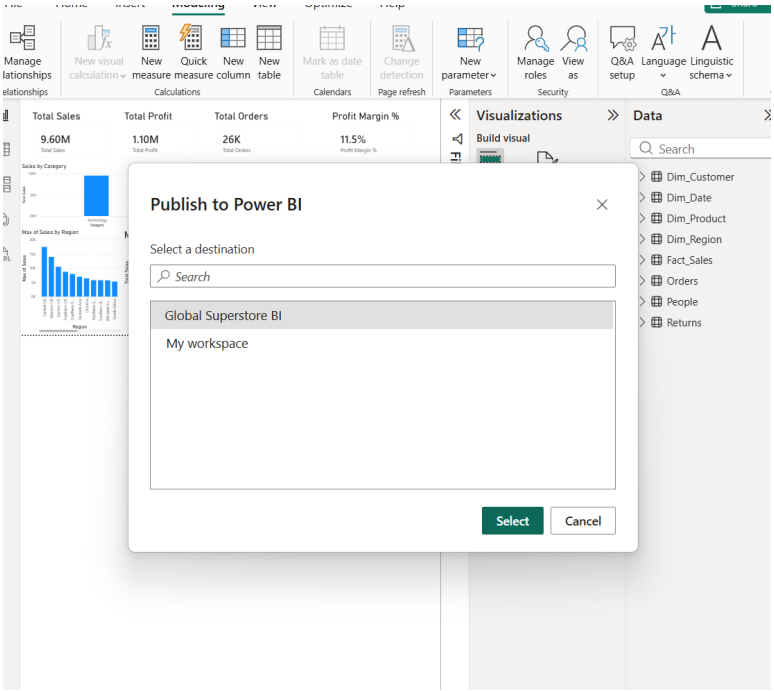


Figure 2: Publishing BI Expe-9 from Power BI Desktop to the Global Superstore BI workspace.

Step 8: Defined permissions used in the access control design.

Permission	Meaning	Power BI mechanism
View	Open and interact with reports, use slicers and filters	Viewer role, report sharing, app audience
Edit	Create and modify reports, datasets and pages	Contributor role or higher
Share	Share reports with others and publish apps	Member role, or 'Allow recipients to share'
Admin	Manage workspace access, roles, settings and deletion	Admin role

Step 9: Assigned permissions to user roles in the Access Control Matrix.

User role	View	Edit	Share	Admin	Workspace role	RLS
CEO	✓	✗	✗	✗	Viewer	None (all data)
Finance Manager	✓	✗	✗	✗	Viewer	None (all data)
Sales Manager	✓	✗	✗	✗	Viewer	None (all data)
Regional Manager	✓	✗	✗	✗	Viewer	Own regions
BI Developer	✓	✓	✗	✗	Contributor	Not applied
BI Administrator	✓	✓	✓	✓	Admin	Not applied

Access was then granted in the workspace using Manage access, then Add people or groups. The student account holds the Admin role and a classmate (DUMISANI CHINYANGA) was added as a Viewer to represent a business user.

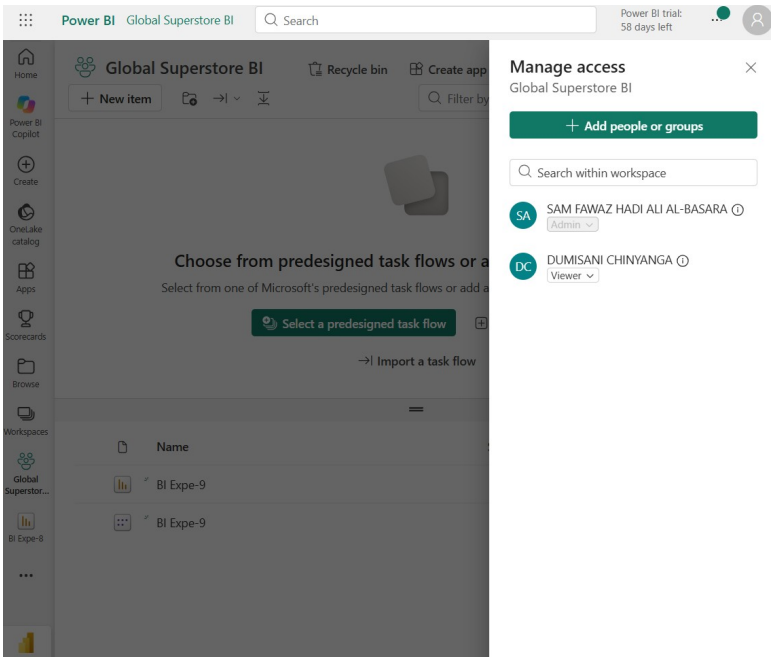


Figure 3: Manage access: Admin (BI Administrator) and Viewer (business user) roles in the Global Superstore BI workspace.

Step 10: Justified permission assignments.

- **Least privilege:** business users (CEO, managers) only consume reports, so they receive View permission and the Viewer role.
- **RLS enforcement:** Regional Managers must be Viewers, because Admin, Member and Contributor roles bypass RLS.
- **Separation of duties:** the BI Developer can edit content but cannot share it or change access, so new content is reviewed before release.
- **Controlled sharing:** only the BI Administrator manages access, which prevents uncontrolled re-sharing of financial data.
- **Need to know:** Finance and the CEO see the Business Scorecard; Regional Managers do not need company-wide targets.

Part D: Row-Level Security (RLS) Planning and Implementation

Step 11: Designed the RLS strategy. The 23 regions were grouped into five management areas, and one static role was created for each area. Each role filters the Fact_Sales table on the Region column, so every visual on every page shows only that area's rows.

RLS role	Regions included	DAX filter on Fact_Sales
Europe Manager	Western, Northern, Southern, Eastern Europe	[Region] IN {"Western Europe", "Northern Europe", "Southern Europe", "Eastern Europe"}
North America Manager	Canada, Central US, Eastern US, Southern US, Western US	[Region] IN {"Canada", "Central US", "Eastern US", "Southern US", "Western US"}
Latin America Manager	Caribbean, Central America, South America	[Region] IN {"Caribbean", "Central America", "South America"}

RLS role	Regions included	DAX filter on Fact_Sales
Africa Manager	Central, Eastern, North, Southern, Western Africa	[Region] IN {"Central Africa", "Eastern Africa", "North Africa", "Southern Africa", "Western Africa"}
Asia Pacific Manager	Central, Eastern, Southeastern, Southern, Western Asia, Oceania	[Region] IN {"Central Asia", "Eastern Asia", "Southeastern Asia", "Southern Asia", "Western Asia", "Oceania"}

Implementation in Power BI Desktop

1. Opened Modeling, then Manage roles, and clicked New.
2. Named the role (for example Asia Pacific Manager) and selected the Fact_Sales table.
3. Switched to the DAX editor, entered the filter expression and validated it.
4. Repeated for all five roles and clicked Save. Power BI confirmed "Successfully applied role changes".

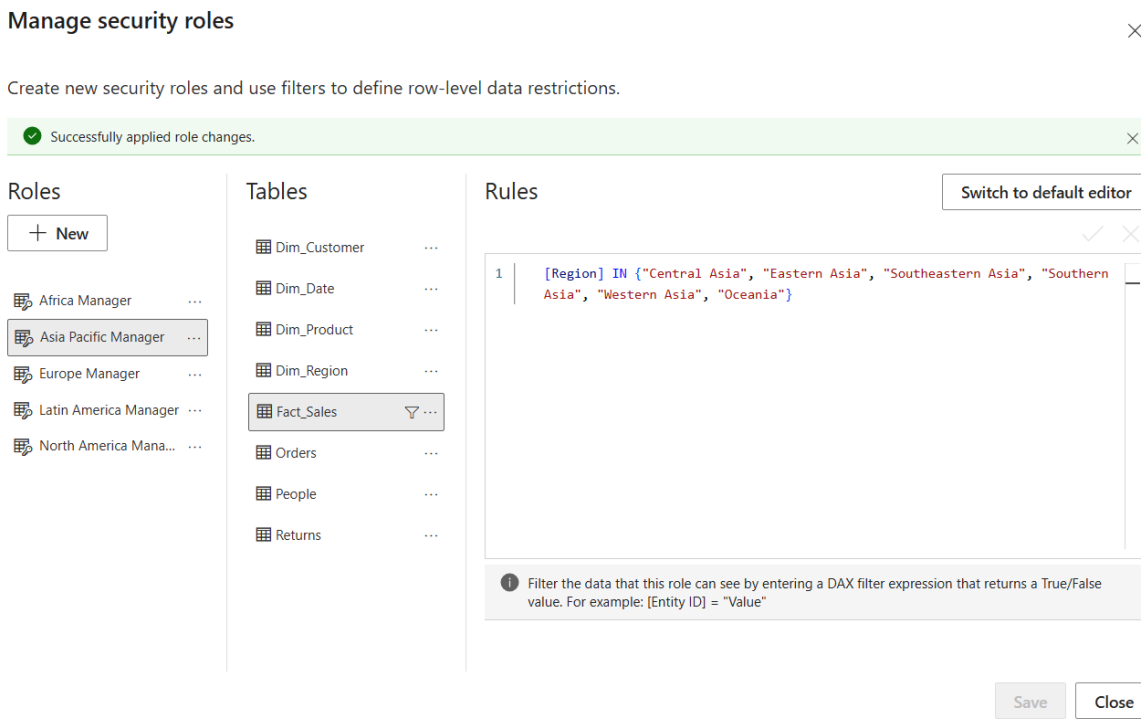


Figure 4: Manage security roles: five regional roles with the Asia Pacific Manager DAX filter on Fact_Sales.

Testing with View as

Each role was tested with Modeling, then View as. The yellow "Now viewing as" bar confirms that the role is active, and the Region slicer and charts show only the regions in that role.

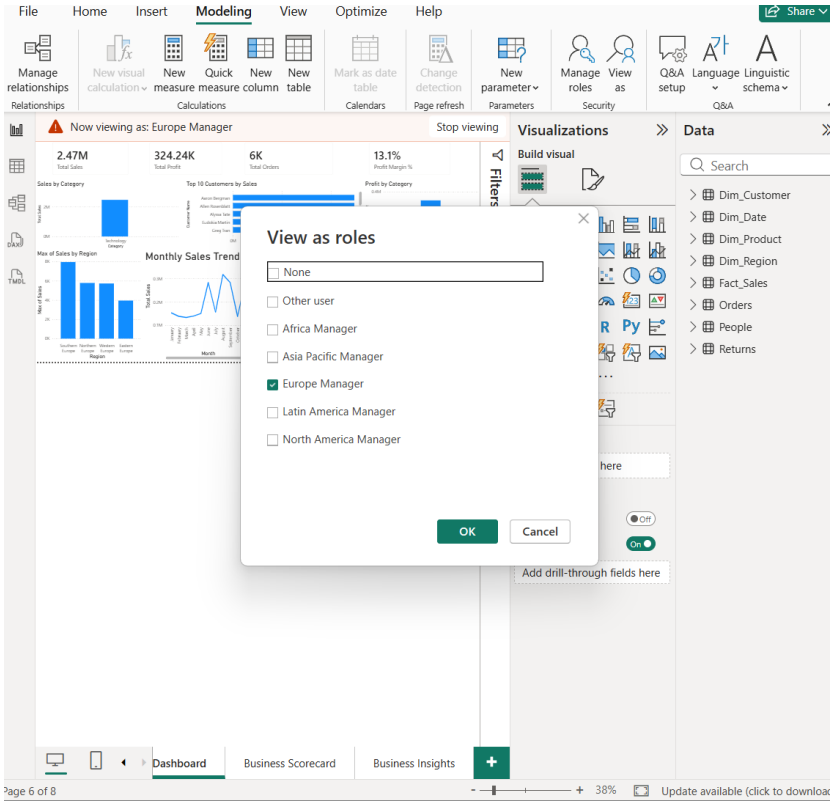


Figure 5: View as Europe Manager: only the four European regions are visible (Sales 2.47M, Profit 324.24K, Margin 13.1%).

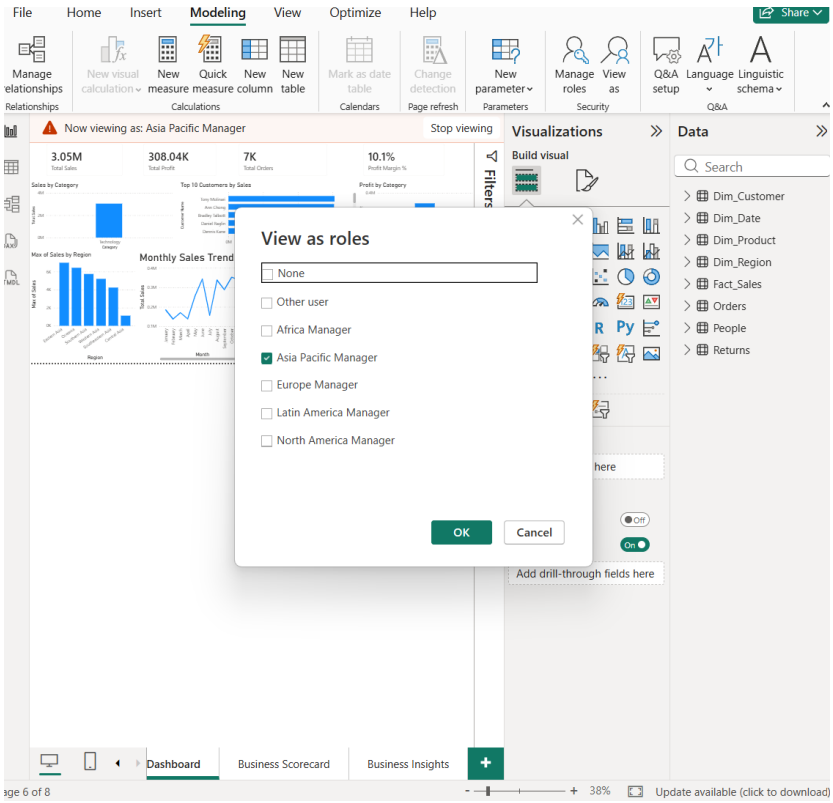


Figure 6: View as Asia Pacific Manager: only Asian regions and Oceania are visible (Sales 3.05M, Profit 308.04K, Margin 10.1%).

View	Total Sales	Total Profit	Total Orders	Profit Margin	Regions visible
No role (full data)	9.60M	1.10M	26K	11.5%	All 23
Europe Manager	2.47M	324.24K	6K	13.1%	4 European regions
Asia Pacific Manager	3.05M	308.04K	7K	10.1%	5 Asian regions and Oceania

The KPI values drop from the company total to each area's own figures, which proves that RLS filters every visual, including the KPI cards, the regional charts and the slicer list.

Assigning roles in the Power BI Service

After publishing, role membership was managed in the Power BI Service: on the BI Expe-9 semantic model, More options, then Security. Members were added to the Europe Manager role and saved.

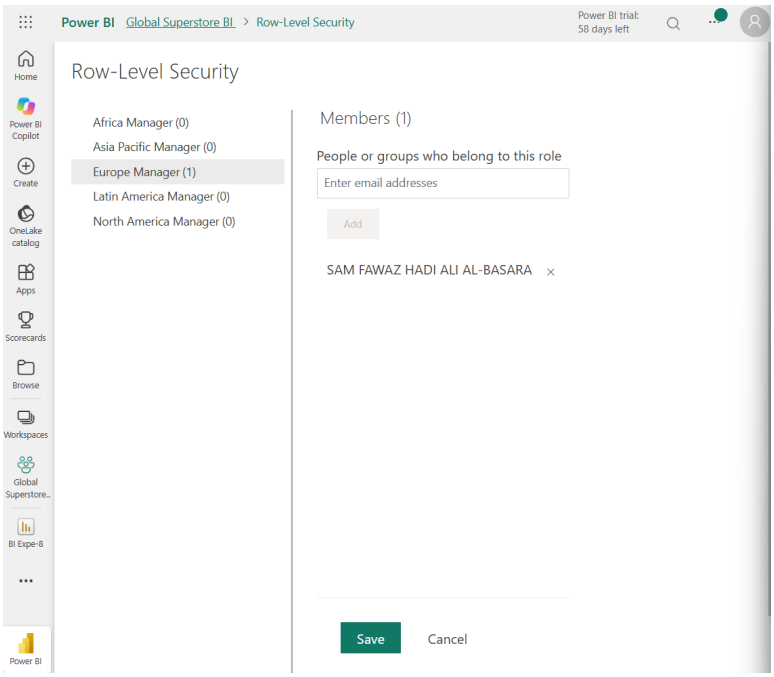


Figure 7: Row-Level Security in the Power BI Service: all five roles published, with a member assigned to Europe Manager.

The role was verified in the Service with Test as role. The report opened with "Now viewing as: Europe Manager" and showed the same filtered results as in Desktop (2.47M sales, four European regions).

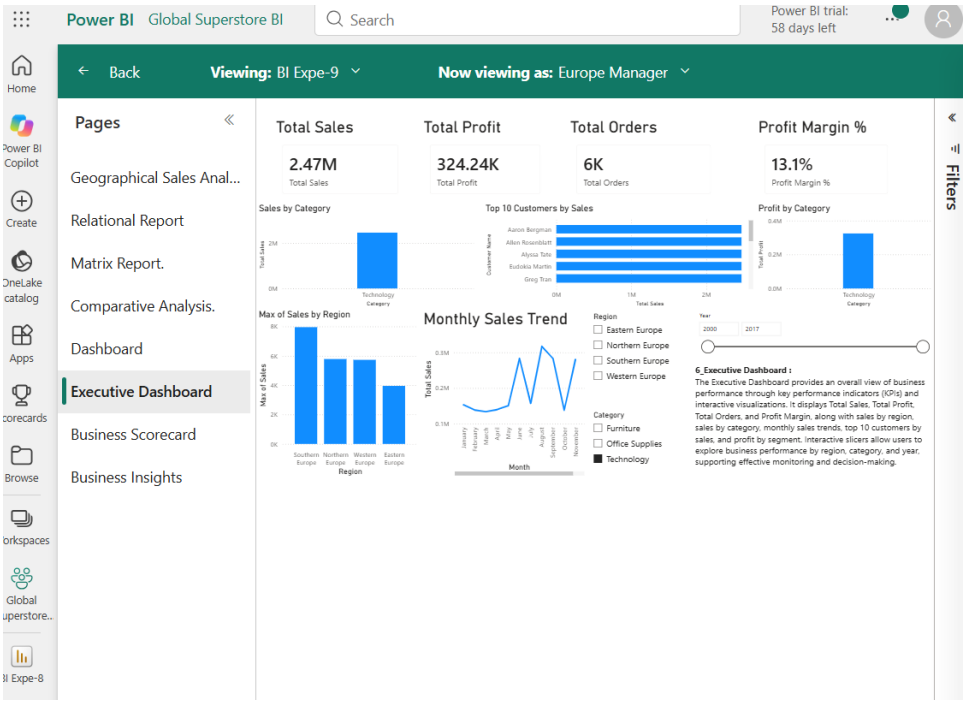


Figure 8: Test as role in the Power BI Service: BI Expe-9 viewed as Europe Manager.

Note: the account added to the Europe Manager role in Figure 7 is the workspace Admin, used for demonstration and testing. Because Admins bypass RLS, in production each Regional Manager would be added to their role with the Viewer workspace role, as designed in the Access Control Matrix.

Dynamic RLS design (scalable option)

For a larger organization, the five static roles can be replaced by one dynamic role. A UserSecurity table (Email, Region) is maintained in Excel, or the existing People table is used, and one role is created with:

```
[Region] IN CALCULATETABLE(VALUES(UserSecurity[Region]),
    UserSecurity[Email] = USERPRINCIPALNAME())
```

New managers are then added by editing the Excel table, without changing DAX or republishing roles.

Step 12: Documented how RLS improves data security.

- **Need-to-know access:** each Regional Manager sees only their own regions' sales, profit and customers.
- **One report, many audiences:** a single report and semantic model serves all managers, so there is no need to maintain separate copies per region.
- **Enforced at data level:** the filter is applied in the semantic model, so it cannot be removed by clearing slicers, changing filters or exporting data.
- **Reduced data leakage:** exports and Analyze in Excel return only the rows the user is allowed to see.
- **Easy to maintain:** access changes are made by adding or removing role members or security group members.

Part E: Security Risk Assessment

Step 13: Identified potential security risks.

Risk	Example in Global Superstore BI	Likelihood	Impact
Unauthorized access	A report link is forwarded to someone outside the intended audience, or a report is published to web	Medium	High
Excessive permissions	A Regional Manager is given Member or Contributor, which bypasses RLS and exposes all regions	Medium	High
Data leakage	Customer names from the Relational Report are exported to Excel and emailed externally	Medium	High
Weak authentication	Password-only sign-in; a phished password gives full access	Medium	High
Stale accounts	A former employee or student still has workspace access	Medium	Medium
RLS misconfiguration	A region name typed wrongly in DAX, so a manager sees no data or the wrong data	Low	Medium

Step 14: Recommended security controls and mitigation measures.

Risk	Control / mitigation
Unauthorized access	Share only with named users or security groups; disable Publish to web in the admin portal; distribute content through a Power BI app
Excessive permissions	Apply least privilege: Viewer for all consumers; review workspace roles quarterly
Data leakage	Restrict export of underlying data; apply sensitivity labels (for example Confidential); keep customer-level pages for authorized roles only
Weak authentication	Enforce MFA through Conditional Access for all users, and always for admins
Stale accounts	Assign access through Entra ID security groups; remove users from groups when roles change
RLS misconfiguration	Test every role with View as and Test as role before release; keep the RLS design document updated

Step 15: Prepared the Security Recommendations Sheet.

No.	Recommendation	Priority	Owner
1	Keep all business users on the Viewer role; never grant Member or Contributor to report consumers	High	BI Administrator
2	Enable RLS for all regional roles and assign members through security groups	High	BI Administrator
3	Enforce Multi-Factor Authentication for every Power BI user	High	IT Security
4	Disable Publish to web and restrict data export for sensitive reports	High	Power BI Admin
5	Apply sensitivity labels to reports containing customer and profit data	Medium	Data Owner
6	Distribute reports through a Power BI app with separate audiences for each department	Medium	BI Administrator
7	Review workspace access and RLS membership every quarter and monitor the activity log	Medium	BI Administrator
8	Move to dynamic RLS with a UserSecurity table as the number of managers grows	Low	BI Developer

Part F: Summary of Deliverables

Deliverable	Where in this report
Role and Permission Matrix	Part A, Step 3 (Role Matrix) and Part C, Step 9
Access Control Matrix	Part C, Steps 8 and 9
Row-Level Security Design Document	Part D, Steps 11 and 12, Figures 4 to 8

Deliverable	Where in this report
Security Risk Assessment Report	Part E, Steps 13 and 14
Security Recommendations Sheet	Part E, Step 15

7. RESULT

A secure BI access strategy was successfully designed and implemented for Global Superstore using authentication, authorization, and access control principles in Microsoft Power BI. Organizational users and their information needs were identified and mapped in a Role Matrix, authentication methods were compared, and an Access Control Matrix was designed using View, Edit, Share and Admin permissions.

A dedicated Global Superstore BI workspace was created with Admin and Viewer roles. Five Row-Level Security roles were implemented on the Fact_Sales table and verified: the Europe Manager role showed only European data (2.47M sales) and the Asia Pacific Manager role showed only Asia Pacific data (3.05M sales), compared with 9.60M for the full company, both in Power BI Desktop and in the Power BI Service. Security risks were assessed and recommendations were prepared to protect organizational data and ensure controlled access.

8. LEARNING OUTCOMES

- Explain authentication, authorization, and access control concepts in BI systems. (CO1-BT2)
- Identify security requirements for different organizational roles. (CO1-BT2)
- Analyze access control mechanisms and permission structures. (CO4-BT4)
- Evaluate security risks associated with BI deployments. (CO4-BT4)
- Design secure access strategies for BI environments. (CO4-BT4)